ISS 2103 • EXAM REVISION PACK 06

Routing, Congestion Control, QoS, SDN & WANs

Course outline weeks 5–7: point-to-point networks, routing, congestion control, internetworking Priority: **HIGH**

In 2023 a **whole 25-mark question** (Q2) was routing and QoS, and every paper since has had at least one 5–10 mark part from this pack. The questions are mostly "explain / compare / tabulate", so learn the tables, the two algorithms and the diagrams.

How often this topic is examined

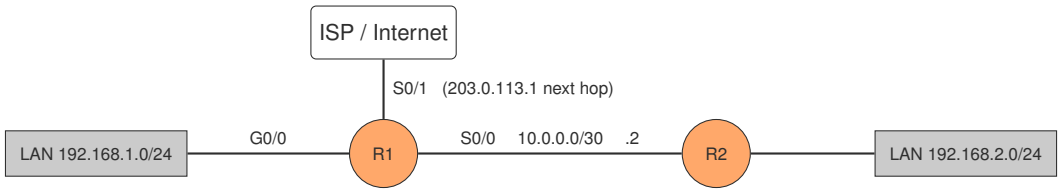
Question asked	Paper(s)	Marks
Three sources of routing-table information	2023 Q2a	3
Link-state vs distance-vector (tabulate)	2023 Q2b	6
Complete Table 1: RIPv1 / EIGRP / OSPF, type & metric	2023 Q2c	6
How EIGRP works and selects the best path	2023 Q2d	5
How RIP v2 works	ISA2024 Q5c	6
Congestion control + two mechanisms	2025 Q3c	10
"Best effort network"	2023 Q3d	4
Three QoS parameters	2023 Q3e	6
QoS of VoIP vs video	2023 Q2e	5
Network performance optimisation (scenario)	ISA2024 Q4d	7
Software-defined networking (SDN)	ISA2024 Q4b	5
Hierarchical structure of the Internet (ISPs)	2018 Q4c	6
SLA: purpose and contents	2023 Q5d.i, Q5d.ii	1 + 2
Two WAN protocols	SE2024 Q1c	2

EXAM TIP

Dijkstra and distance-vector table updates have *not* been asked yet but sit squarely in the outline ("routing algorithms"). A 6–10 mark "find the shortest paths" question is a real risk, so practise Sections 3 and 4.

1 Routing basics and the routing table

Routing is the act of moving information across an internetwork from a source to a destination. It involves two activities: **determining optimal paths** (the routing algorithm) and **forwarding packets** along them (switching). A router reads the **destination IP address**, looks for the longest matching prefix in its **routing table** and sends the packet out of the matching interface to the next hop.



Code	Network	[AD/metric] next hop	Interface
C	192.168.1.0/24	directly connected	G0/0
C	10.0.0.0/30	directly connected	S0/0
O	192.168.2.0/24	[110/65] via 10.0.0.2	S0/0
S*	0.0.0.0/0	[1/0] via 203.0.113.1	S0/1

Figure: R1's routing table holds all three sources: connected (C), dynamic (O = OSPF) and static (S, here the default route).

PAST PAPER **2023 Q2a** **[3 marks]**

Natalie intends to setup a network with several routers purchased by the organization she is working for. She has however, forgotten how routing tables work and needs this information to decide the routing protocol to use. To aid her, describe three (3) different sources a router can use to put/maintain information contained in a routing table.

Model answer

1. **Directly connected networks**: when an interface has an IP address and is up, the router adds that network automatically (code C, AD 0).
2. **Static routes**: entered manually by the administrator, e.g. `ip route 192.168.2.0 255.255.255.0 10.0.0.2` (code S, AD 1). Includes the **default route** 0.0.0.0/0 for "everything else".
3. **Dynamic routing protocols**: routes learned automatically from neighbouring routers through RIP, OSPF, EIGRP or BGP; they adapt when the topology changes (codes R, O, D).

If two sources offer the same network, the router keeps the one with the lowest **administrative distance**.

1.1 Static vs dynamic routing

Administrative distance = how much a source is trusted. Connected 0, static 1, EIGRP 90, OSPF 110, RIP 120. Lower wins.

The lecturer's notes list three classes of routing algorithm: **flooding** (send every packet out of every line except the one it came in on: robust but wasteful, needs a hop counter), **static** and **dynamic (adaptive)**.

	Static (non-adaptive)	Dynamic (adaptive)
How routes appear	Table mappings typed in by the administrator	Learned from routing update messages
Reaction to change	None until the admin edits it	Recalculates and re-advertises automatically
Overhead	No CPU or bandwidth used	Uses CPU, memory and link bandwidth
Security	Nothing advertised, so more secure	Updates can be spoofed (use authentication)
Best for	Small, stable networks; stub networks; default route	Large, changing networks with redundant paths

Design goals of a routing algorithm (lecturer's list)

Optimality (picks the best route by its metric), **simplicity and low overhead**, **robustness and stability**, **rapid convergence** (all routers agree on optimal routes quickly; slow convergence causes loops), **flexibility**. **Metrics** used: path length / hop count, reliability, delay, bandwidth, load, communication cost.

2 Distance vector vs link state

PAST PAPER **2023 Q2b** **[6 marks]**

Tabulate at least three (3) main differences between link-state and distance-vector routing protocols.

"Tabulate" means a table. Three rows earns 6 marks (2 each: one fact per column). Write 4 rows to be safe.

Model answer

Point	Distance vector	Link state
View of network	Knows only distance and direction (next hop) to each network: "routing by rumour"	Every router builds a full map (topology database) of the network
What is sent	Whole (or part of) routing table	Small link-state advertisements (LSAs) describing its own links
Sent to	Neighbours only	All routers in the area (flooding)
When	Periodically (RIP every 30 s)	Only when a link changes (+ small hellos)
Algorithm	Bellman-Ford	Dijkstra shortest path first (SPF)
Convergence	Slow; routing loops, count-to-infinity	Fast; loop-free
Resources	Little CPU and memory	More CPU and memory (LSDB + SPF)
Scale / examples	Small networks: RIPv1, RIPv2, IGRP	Large hierarchical networks (areas): OSPF, IS-IS

PAST PAPER **2023 Q2c** **[6 marks]**

Copy and complete Table 1 below;

Routing Protocol	Distance-vector or Link-state?	Metric used
RIPv1		
EIGRP		
OSPF		

Table 1: Routing

Model answer

Routing protocol	Distance-vector or link-state?	Metric used
RIPv1	Distance-vector (classful)	Hop count (max 15; 16 = unreachable)
EIGRP	Advanced distance-vector (often called hybrid)	Composite: bandwidth (slowest link) + delay (sum) by default; load and reliability optional
OSPF	Link-state	Cost $= \frac{\text{reference bandwidth } (10^8 \text{ bps})}{\text{interface bandwidth}}$, summed along the path

TRAP – where students lose marks

Writing "EIGRP = link-state". EIGRP is a Cisco **distance-vector** protocol with some link-state features (hellos, partial updates); if you write "hybrid", also say "advanced distance vector".

3 Link state: Dijkstra's shortest path worked example

Each link-state router floods LSAs, builds the same **link-state database**, then runs **Dijkstra** with itself as the root to get a **shortest-path tree**. Method:

1. Put the source in the set N' (done). Every neighbour gets its link cost $D(v)$; others get ∞.
2. Pick the node not in N' with the **smallest** D; add it to N'.
3. For each neighbour v of that node: $D(v) = \min\big(D(v),\ D(w)+c(w,v)\big)$, and record the predecessor $p(v)$.
4. Repeat until all nodes are in N'. Follow predecessors back to read each path.

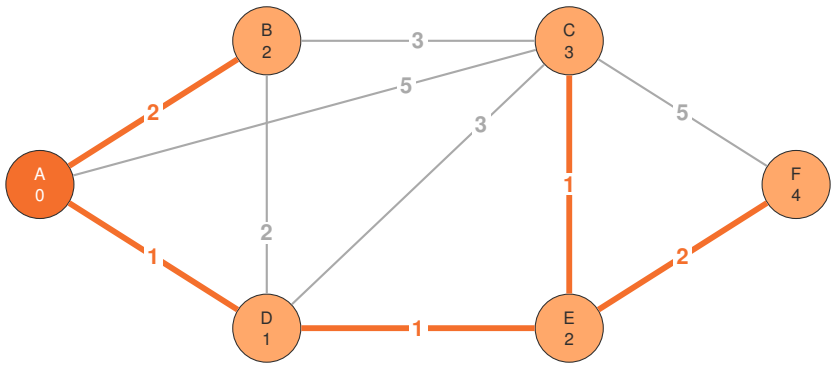


Figure: weighted graph, source A. Thick orange = shortest-path tree; the number in each node is its final cost from A.

Dijkstra from A: D(v), p(v) after each step

Step	N'	B	C	D	E	F
0	A	2, A	5, A	**1, A**	∞	∞
1	A D	**2, A**	4, D		2, D	∞
2	A D B		4, D		**2, D**	∞
3	A D B E		**3, E**			4, E
4	A D B E C					**4, E**
5	A D B E C F					

Step 1: $D(C) = \min(5,\ 1+3) = 4$ via D ← *via D beats the direct link*
$D(E) = \min(\infty,\ 1+1) = 2$ via D
Step 2: B, E tie on 2; take B ← *$D(C)$ via $B = 5 > 4$: no change*
Step 3: $D(C) = \min(4,\ 2+1) = 3$ via E
$D(F) = \min(\infty,\ 2+2) = 4$ via E
Step 4: $D(F) = \min(4,\ 3+5) = 4$ ← *no change*

Destination	Shortest path	Next hop (forwarding table of A)	Cost
B	A–B	B	2
C	A–D–E–C	D	3
D	A–D	D	1
E	A–D–E	D	2
F	A–D–E–F	D	4

EXAM TIP

In the exam: (1) redraw the graph, (2) build the step table with "cost, predecessor" in every cell, (3) show one min(. . .) line for each change, (4) finish with the path/next-hop table and **highlight the shortest-path tree** on your graph. Ties can be broken either way; say which you chose.

Bold entry in each row = the node added to N' in the next step. Once a node is in N', its cost is final; leave its column blank.

4 Distance vector: Bellman-Ford worked example

Each router keeps a **distance vector** (best known cost to every destination) and sends it to its **neighbours**. On receiving vectors it applies the **Bellman-Ford equation**:

$$D_x(y) = \min_{v \in \text{neighbours}} \{ c(x, v) + D_v(y) \}$$

"my cost to y = cost to neighbour v + what v says its cost to y is", taking the cheapest neighbour.

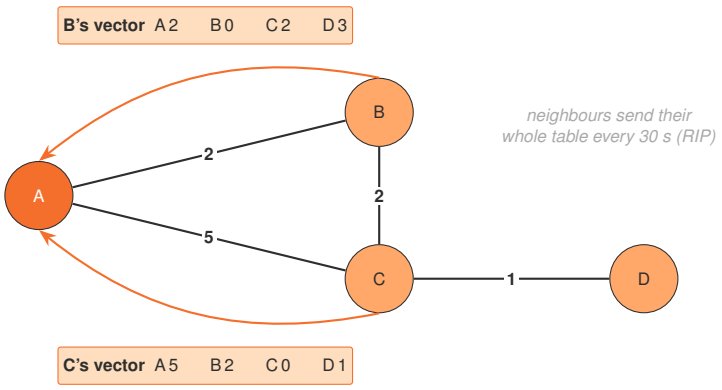


Figure: A receives the distance vectors of its neighbours B (link cost 2) and C (link cost 5).

A's new table from the two vectors

$D_A(B) = \min(2+0,\ 5+2) = 2$ via B

$D_A(C) = \min(2+2,\ 5+0) = 4$ via B ← *the direct link (5) is not the best*

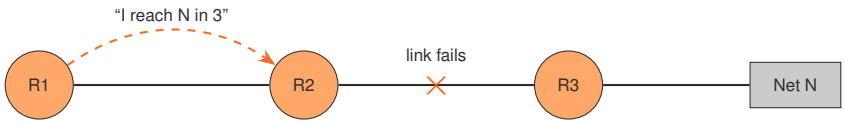
$D_A(D) = \min(2+3,\ 5+1) = 5$ via B ← *$c(A,B) + D_B(D)$ vs $c(A,C) + D_C(D)$*

Dest	via B $(2 + D_B)$	via C $(5 + D_C)$	A's entry (cost, next hop)
B	2	7	**2, B**
C	4	5	**4, B**
D	5	6	**5, B**

A then sends its *new* vector to B and C. This repeats until no table changes: the network has **converged**.

4.1 Count-to-infinity and its fixes

Good news (a cheaper route) spreads fast; **bad news (a failure) spreads slowly**, because routers keep believing each other's old routes.



R2 has lost N but believes R1, not knowing R1's route goes through R2 itself

Hop counts to Net N after the failure (RIP, cost 1 per hop)

Exchange	before	1	2	3	4	…	final
R2	2	4 (via R1)	4	6	6	…	16
R1	3	3	5 (via R2)	5	7	…	16

$D_{R2}(N) = 1 + D_{R1}(N) = 1 + 3 = 4$ ← *R1's stale route really goes through R2 itself*

$D_{R1}(N) = 1 + D_{R2}(N) = 5$, then $6, 7, \ldots$ ← *counts up until 16 = "infinity"*

Fixes (each stops the loop a different way):

- **Maximum hop count**: define infinity as 16 (RIP), so counting stops quickly.
- **Split horizon**: never advertise a route back out of the interface it was learned on (R1 would not tell R2 about N).
- **Route poisoning**: the router that loses a route advertises it at once with metric 16 (unreachable).
- **Poison reverse**: neighbours echo the poisoned route back with metric 16 to confirm it.
- **Hold-down timer**: after a route goes down, ignore worse news about it for 180 s so false routes cannot be accepted.

- **Triggered updates**: send an update immediately on a change instead of waiting for the 30 s timer.

5 Routing protocols: RIP v2, EIGRP and OSPF

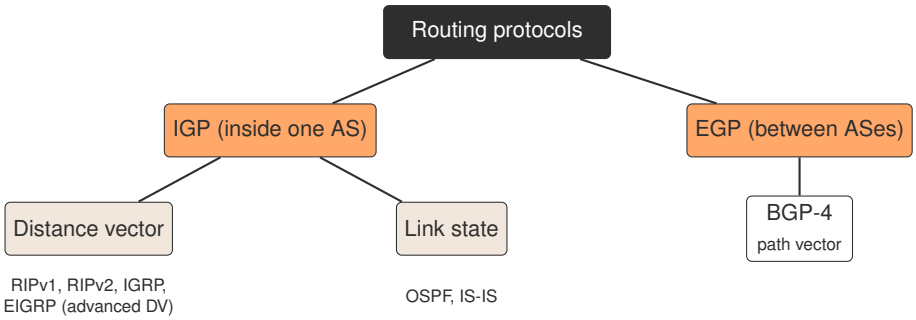


Figure: AS = autonomous system, one organisation's routing domain. BGP connects ISPs.

PAST PAPER **ISA2024 Q5c** **[6 marks]**

Explain in detail the working of the RIP v2 routing algorithm.

Model answer

1. RIPv2 is a **distance-vector** protocol using the **Bellman-Ford** algorithm; metric = **hop count**, max 15 (16 = unreachable), so it suits small networks. AD 120, UDP port 520.
2. Each router starts with its directly connected networks (hop 0) and every **30 s** sends its **entire routing table** to neighbours using multicast **224.0.0.9**.
3. On receiving an update it adds **1 hop** to every route; a new route, or one with fewer hops, is installed with that neighbour as next hop.
4. **Classless**: sends the **subnet mask** with every route, so it supports VLSM and CIDR (RIPv1 did not); also adds **authentication** (plain text / MD5), a next-hop field and route tags.
5. **Timers**: update 30 s, invalid 180 s (route marked unreachable), hold-down 180 s, flush 240 s (removed).
6. **Loop prevention**: split horizon, route poisoning, poison reverse, hold-down timers and triggered updates.

Weakness: slow convergence; hop count ignores bandwidth (a 2-hop 64 kbps path beats a 3-hop 1 Gbps path).

PAST PAPER **2023 Q2d** **[5 marks]**

Describe the workings of the EIGRP routing protocol, clearly highlighting how it selects the best path for transmission of packets.

RIPv1 vs v2 in one line: v1 = classful, broadcast 255.255.255.255, no auth; v2 = classless, multicast 224.0.0.9, MD5 auth.

Model answer

1. **EIGRP** (Enhanced Interior Gateway Routing Protocol) is Cisco's **advanced distance-vector** protocol: classless, AD 90, protocol number 88.
2. Routers discover neighbours with **Hello** packets (multicast 224.0.0.10) and keep three tables: **neighbour**, **topology** (all routes from all neighbours) and **routing** (best routes).
3. Updates are **partial and bounded**: only changes, only to routers that need them, sent reliably with RTP. This saves bandwidth.
4. **Metric**: composite of the **slowest bandwidth** on the path and the **sum of delays**: metric $= 256 \times (10^7/BW_{\text{min}} + \text{delay}/10)$, with BW in kbps and delay in μs. Lowest metric = best.
5. **Best path by DUAL**: the route with the lowest **feasible distance (FD)** becomes the **successor** and goes into the routing table. A backup whose **reported distance (RD)** $<$ **FD** of the successor (feasibility condition) is kept as the **feasible successor**. If the successor fails, the feasible successor is used at once: fast, loop-free convergence. EIGRP also allows

unequal-cost load balancing (`variance`).

OSPF in six lines (link state)

- Open standard link-state IGP, AD 110, IP protocol 89; uses Dijkstra SPF.
- Hellos (224.0.0.5) form **adjacencies**; on a LAN a **DR**/**BDR** is elected to cut LSA traffic.
- Routers flood **LSAs**; every router in an area has an identical **LSDB**, then runs SPF to build its tree.
- Hierarchical **areas**: all areas attach to backbone **area 0**; this keeps LSDBs and SPF runs small.
- Metric = **cost** = 10^8/bandwidth: 100 Mbps = 1, 10 Mbps = 10, T1 1.544 Mbps = 64.
- Fast convergence, classless (VLSM), authentication, equal-cost load balancing.

6 Congestion control

Congestion happens when the load (packets offered) is greater than the capacity of part of the network. Router queues fill, delay rises, packets are dropped, senders retransmit, and the extra traffic makes it worse (**congestion collapse**). **Congestion control** = techniques that keep the load below capacity: **prevent** congestion (open loop) or **remove** it once it is detected (closed loop). Causes: bursty traffic, slow links/routers, too little buffer space, many inputs feeding one output.

Open loop (prevention, before it happens)	Closed loop (removal, after it is detected)
Retransmission policy: sensible timers	Backpressure: congested node refuses packets, upstream nodes slow, back to the source
Window policy: Selective Repeat, not Go-Back-N	Choke packet: router tells the source directly to slow down (ICMP source quench)
Acknowledgement policy: cumulative / delayed ACKs	Implicit signalling: source infers it from lost ACKs / long delay (TCP)
Discard policy: drop the less important packets first	Explicit signalling: routers mark packets (ECN, Frame Relay FECN/BECN)
Admission control: refuse a new flow the network cannot carry	RED: drop/mark randomly *before* the queue is full
Traffic shaping: leaky bucket, token bucket	TCP: slow start, AIMD, halve the window on loss

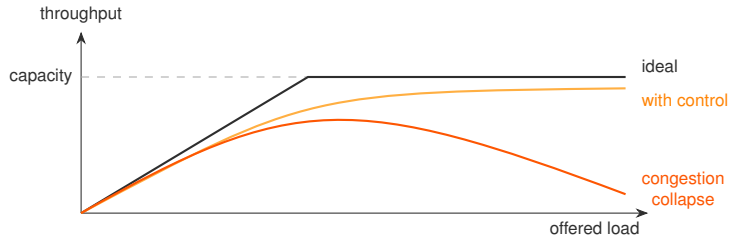


Figure: without congestion control, throughput falls as load rises past capacity (retransmissions and drops).

6.1 Traffic shaping: leaky bucket and token bucket

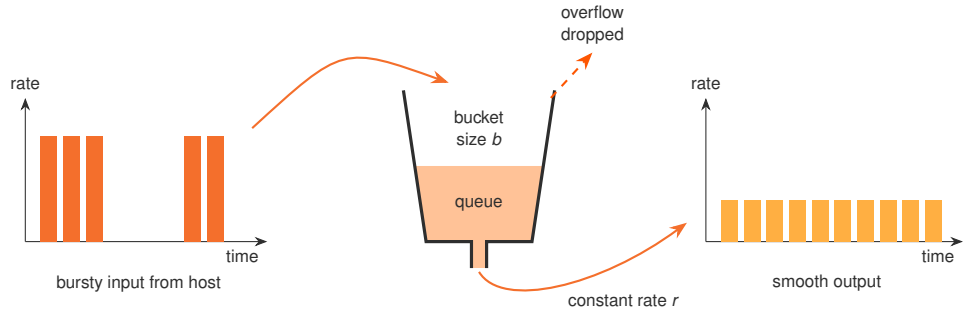


Figure: leaky bucket. Water poured in at any rate leaks out of the hole at a fixed rate; if the bucket is full, new packets are discarded.

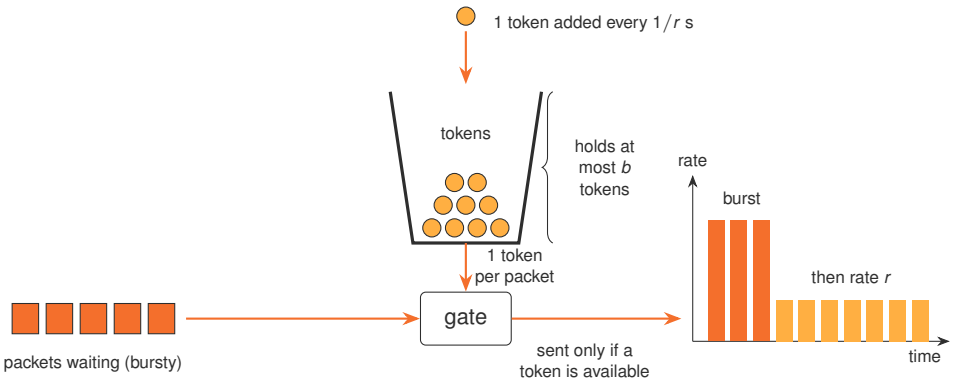


Figure: token bucket. Saved-up tokens let a burst of up to b packets go at once; after that the output falls to the token rate r.

	Leaky bucket	Token bucket
Output	Always the constant rate r	Bursts up to b, average rate r
Idle host	Saves nothing	Saves tokens (up to b) for later bursts
When full	Discards **packets**	Discards **tokens**, never packets
Use	Strict smoothing (e.g. ATM)	Flexible shaping/policing of bursty data

Token bucket: how long can a full burst last?

$$b + rS = MS \qquad \leftarrow \textit{saved tokens + new tokens = data sent at max rate } M$$

$$S = \frac{b}{M - r}$$

$$S = \frac{250\ \text{KB}}{(25 - 2)\ \text{MB/s}} \approx 10.9\ \text{ms} \qquad \leftarrow \textit{e.g. } b = 250\ \text{KB},\ r = 2,\ M = 25\ \text{MB/s}$$

PAST PAPER **2025 Q3c** **[10 marks]**

A network is experiencing a high volume of traffic, leading to delays and dropped packets. Explain the concept of congestion control and describe at least two mechanisms used to manage network congestion.

Mnemonic: **leaky** = constant drip, drops **packets**. **Token** = savings account: you can spend a burst, drops **tokens**.

Model answer

Concept (3): congestion occurs when the traffic offered exceeds the capacity of links/routers. Queues fill, so delay rises and packets are dropped; retransmissions add even more load. **Congestion control** is the set of mechanisms that keep the load below capacity, either **preventing** congestion (open loop) or **detecting and removing** it (closed loop).
Mechanisms (7):

1. **Leaky bucket (traffic shaping)**: packets enter a finite queue and leave at a **constant rate**, so bursts are smoothed; when the bucket is full packets are discarded. (Draw the figure above.)
2. **Token bucket**: tokens arrive at rate r into a bucket of size b; a packet may leave only by taking a token. Allows controlled bursts while limiting the average rate; tokens, not packets, are discarded.
3. **Choke packets**: a congested router sends a control packet to the source, which reduces its sending rate.
4. **Backpressure**: the congested node stops accepting packets from its upstream neighbour, which in turn slows its own upstream, back to the source.
5. **TCP congestion control**: slow start (window doubles each RTT to the threshold), then additive increase; on loss the window is halved (AIMD) or reset to 1.
6. **RED** / **ECN** and **admission control**: routers drop or mark early so senders slow down before queues overflow; new flows are refused if the network cannot carry them.

EXAM TIP

For 10 marks: define (2), cause/effect (1), open vs closed loop (1), then **two mechanisms in detail with a diagram** (3 each). The leaky and token buckets are the easiest to draw.

7 Quality of Service (QoS)

QoS = the ability of the network to give some traffic better, more predictable treatment (bandwidth, delay, jitter, loss) than other traffic. Techniques: **classification and marking** (DSCP in the IP header), **scheduling** (FIFO, priority queuing, weighted fair queuing), **traffic shaping/policing**, **admission control**, **resource reservation**. Two models: **IntServ** (per-flow reservation with RSVP: guaranteed but does not scale) and **DiffServ** (packets marked into a few classes, e.g. EF for voice; each router gives per-hop treatment: scales).

PAST PAPER **2023 Q3d** **[4 marks]**

The Internet is often described as being a “best effort network”. Explain what is meant by the term “best effort network”.

Model answer

1. The network **tries** to deliver every packet but gives **no guarantees**: not of delivery, delay, bandwidth, order or freedom from duplicates.
2. All packets are treated **equally** (first come, first served); there is no reservation or priority.
3. When routers are congested, packets are simply **dropped**; IP is connectionless and does not tell anyone.
4. Reliability, if needed, is left to the end hosts (TCP retransmits). It is simple, cheap and scalable, but unpredictable for real-time traffic, which is why QoS (DiffServ, IntServ, MPLS) was added.

PAST PAPER **2023 Q3e** **[6 marks]**

List three QoS parameters that are often measured to characterize the behavior of a network or network connection.

Model answer

1. **Bandwidth** / **throughput**: amount of data successfully delivered per second (bps).
2. **Latency (delay)**: time for a packet to go from source to destination (propagation + transmission + queuing + processing).
3. **Jitter**: variation in the delay (packet arrival time) between packets of the same flow; harms real-time voice and video.
4. **Packet loss**: percentage of packets that never arrive, mostly from congestion.

(Also: reliability / error rate, availability.)

PAST PAPER **2023 Q2e** **[5 marks]**

Discuss the quality-of-service requirements of a Voice-over-IP (VoIP) application and how they differ from those of a video-based application.

“List three” for 6 marks: name (1) + one-line definition (1) each. Give a fourth in case one is not accepted.

Model answer

Parameter	VoIP	Video
Bandwidth	Low: about 30–100 kbps per call	High: 1–25 Mbps (HD/4K), bursty
Delay	Very sensitive: one-way $<$ 150 ms (interactive)	Streaming tolerant (buffer hides seconds); conferencing $<$ 150–200 ms
Jitter	$<$ 30 ms; jitter buffer used	Streaming tolerant; conferencing sensitive
Loss	$<$ 1%; lost packets are not resent (UDP/RTP)	$<$ 0.1–1%; loss gives pixelation and freezing
Traffic	Small, constant-rate packets, two-way	Large packets, variable rate, mostly one-way
Treatment	Highest priority (DiffServ EF), low-latency queue	Lower class (AF41 conferencing, AF31 streaming)

In short: voice needs **little bandwidth but very low delay and jitter**; video needs **much more bandwidth** and is more tolerant of delay when streamed.

PAST PAPER **ISA2024 Q4d** **[7 marks]**

A large enterprise network is experiencing slow data transfer rates, high latency and frequent packet losses. The network administrator is tasked with improving the network's performance to ensure efficient and reliable communication among employees, customers and partners. Describe network performance optimization and how would you achieve it in this scenario to improve the network's efficiency, speed and reliability?

Model answer

Network performance optimisation = monitoring, analysing and tuning the hardware, configuration and traffic so the network gives the most throughput with the least latency, jitter and loss for business applications.

1. **Monitor and baseline**: SNMP/NetFlow tools and Wireshark to find bottlenecks, top talkers, error counters, overloaded links.
2. **Upgrade bandwidth and hardware**: faster uplinks (1 $\rightarrow$ 10 Gbps), replace hubs with switches, fix faulty cables/NICs (CRC errors cause loss).
3. **Segment**: VLANs and subnetting cut broadcast traffic and collision domains.
4. **QoS**: classify and prioritise VoIP and business apps over bulk downloads; shape/police traffic (token bucket).
5. **Load balancing and redundancy**: link aggregation, redundant routers (HSRP), dynamic routing (OSPF/EIGRP) for fast failover: reliability.
6. **Caching and WAN optimisation**: proxy caches/CDN, compression and deduplication on WAN links.
7. **Security and housekeeping**: block malware and unauthorised streaming, update firmware, document and review regularly.

8 Software-defined networking (SDN)

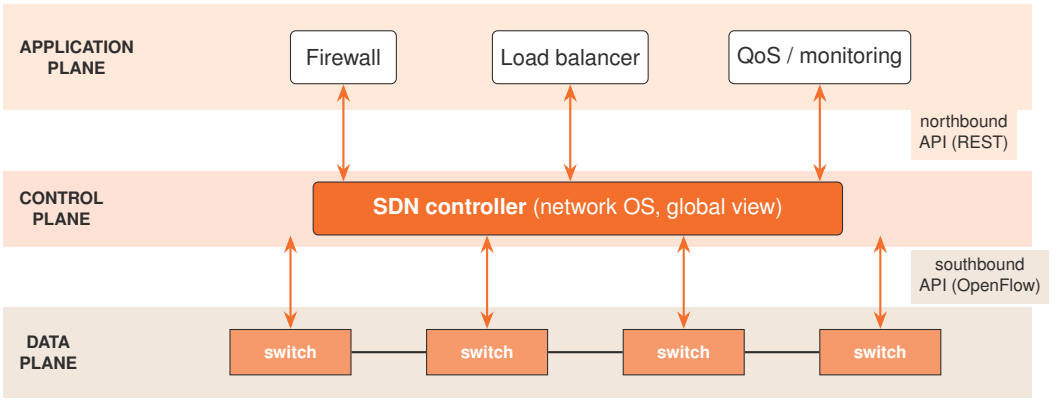


Figure: SDN architecture. The control plane is taken out of each device and centralised in the controller; switches only forward.

PAST PAPER **ISA2024 Q4b** **[5 marks]**

Elaborate on software-defined networking (SDN).

Model answer

1. SDN is an architecture that **separates the control plane** (deciding where traffic goes) **from the data plane** (forwarding packets).
2. Control is **centralised in software**, the **SDN controller** (e.g. OpenDaylight, ONOS), which has a global view of the network; switches become simple forwarding devices.
3. The controller programs the switches' flow tables through a **southbound API** (OpenFlow); applications request services through a **northbound API** (REST).
4. **Benefits**: programmable and automated, central management, fast provisioning, network-wide policies for QoS and security, vendor independence, suits cloud and virtualisation.
5. **Challenges**: the controller is a single point of failure and attack; scalability; new skills needed.

Traditional router: control + data plane in every box, configured one by one. SDN: one brain, many simple "hands".

9 Internet structure: ISPs, tiers and SLAs

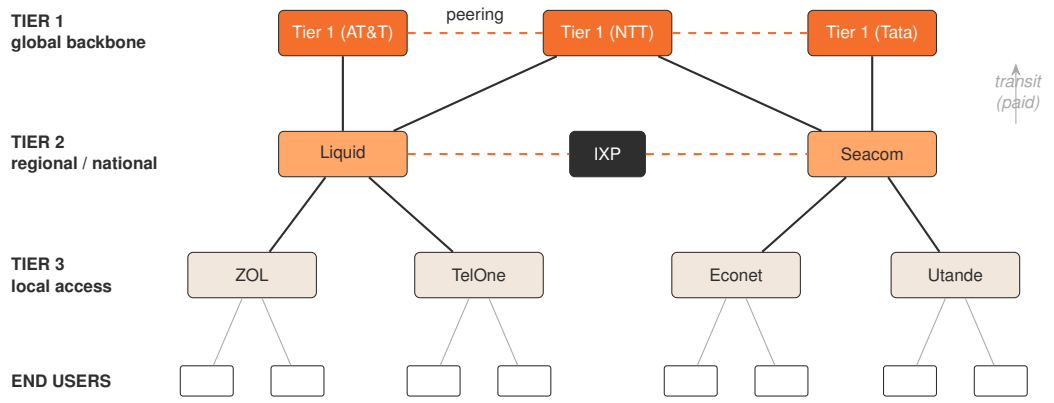


Figure: hierarchical structure of the Internet. Solid = customer buys transit from the tier above; dashed = settlement-free peering.

PAST PAPER **2018 Q4c** **[6 marks]**

Draw and explain the hierarchical structure of the Internet (ISPs).

Model answer

(Diagram above: 2 marks.)

1. **Tier 1 ISPs**: global backbone networks that can reach every network without buying transit; they **peer** with each other free of charge. E.g. AT&T, NTT, Tata, Lumen.
2. **Tier 2 ISPs**: regional / national providers; **buy transit** from Tier 1 and peer with other Tier 2s at **IXPs**. E.g. Liquid Intelligent Technologies, Seacom.
3. **Tier 3 ISPs**: local access ISPs that buy transit from Tier 2 and connect homes and businesses. E.g. ZOL, TelOne, Econet, Utande.
4. Networks meet at **POPs** (points of presence) and **IXPs** (Internet exchange points); content providers (Google, Netflix) connect directly to cut cost. Traffic goes up the hierarchy and back down; each ISP is an autonomous system and they exchange routes with **BGP**.

PAST PAPER **2023 Q5d.i / Q5d.ii** **[1 + 2 marks]**

Telecommunication companies (Telcos) that provide global network services, define the services they offer to customers within a Service Level Agreement (SLA). (i) Explain the purpose of a SLA. (ii) Give any two (2) examples of what it might contain.

Model answer

(i) An SLA is a **formal contract between a service provider and a customer** that defines the level and quality of service to be delivered, how it is measured, and the penalties if it is not met. (ii) Any two: guaranteed **availability/uptime** (e.g. 99.9%); guaranteed **bandwidth**; maximum **latency, jitter and packet loss**; **fault response and repair times**; support hours and contacts; **penalties** / **service credits**; reporting and monitoring; security responsibilities.

What 99.9% uptime really allows

$$(1 - 0.999) \times 365 \times 24 = 8.76 \text{ h of downtime per year} \quad \leftarrow \textit{99.99\% allows only 53 min}$$

10 WAN technologies and protocols

A **WAN** connects LANs over a large area using a carrier's (telco's) links. The customer's router is the **DTE**; the CSU/DSU or modem is the **DCE** that clocks the line. The **local loop** runs from the customer's **demarcation point** to the provider's exchange. These are **point-to-point networks**: data crosses intermediate switching nodes (store and forward) using circuit switching, datagrams or **virtual circuits**.

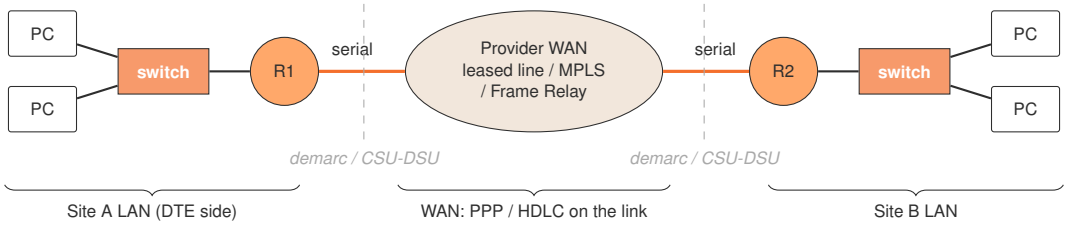


Figure: a WAN link between two sites. The routers' serial interfaces run a layer 2 WAN protocol over the carrier's service.

- **Leased line**: Dedicated, always-on point-to-point circuit rented from the telco (E1 = 2.048 Mbps, T1 = 1.544 Mbps). Guaranteed bandwidth and private, but expensive and fixed.
- **PPP**: Layer 2 protocol for direct links (serial, DSL as PPPoE). LCP sets up the link, NCPs carry IP and others; authentication PAP / CHAP; error detection.
- **HDLC**: ISO bit-oriented synchronous protocol; flag 01111110 with bit stuffing; Cisco default on serial links (proprietary version). No authentication.
- **X.25**: Old packet-switched VC network; error checking at *every* hop, so reliable on poor lines but slow (64 kbps).
- **Frame Relay**: Packet-switched VCs (PVCs identified by DLCIs); no hop-by-hop error correction, so faster than X.25; CIR; congestion bits FECN/BECN; now legacy.

- **ATM**: Fixed 53-byte cells (5 header + 48 data); VCs; QoS classes for voice and video.
- **MPLS**: Routers forward on a short **label** instead of an IP lookup; label-switched paths; fast, supports QoS, traffic engineering and VPNs. Today's carrier standard.
- **Broadband / VPN**: DSL, cable, 4G/5G, fibre with an IPsec VPN over the Internet: cheap but best effort.

PAST PAPER **SE2024 Q1c** **[2 marks]**

Explain two WAN protocols.

Model answer

1. **PPP (Point-to-Point Protocol)**: a layer 2 protocol for direct links between two nodes (serial lines, PPPoE on DSL). It gives framing, error detection, link set-up with LCP, multi-protocol support with NCPs and authentication with PAP/CHAP.
2. **HDLC (High-level Data Link Control)**: a bit-oriented synchronous data-link protocol for point-to-point links, using the flag 01111110 and bit stuffing; the default on Cisco serial interfaces.

(Also accepted: Frame Relay, MPLS, X.25, ATM.)

Two marks, two protocols: name + one solid sentence each. Do not write a page.

11 Practice questions

PRACTICE Q1 EASY **[4 marks]**

Distinguish between static and dynamic routing, giving one situation where each is the better choice.

Answer

Static: routes typed in by the administrator; no overhead and secure, but do not adapt to failures. Best for a small branch with one link to head office (a stub network with a default route). **Dynamic**: routers learn routes from each other with a protocol (OSPF, EIGRP) and recalculate when links fail; uses CPU and bandwidth. Best for a large network with redundant paths.

PRACTICE Q2 EASY **[3 marks]**

A router learns a route to 10.1.1.0/24 from OSPF, from RIP and from a static route. Which one is placed in the routing table? Explain.

Answer

The **static route**. The router prefers the source with the lowest **administrative distance**: static 1 $<$ OSPF 110 $<$ RIP 120. Metrics are only compared between routes from the *same* protocol.

PRACTICE Q3 EASY **[3 marks]**

Define latency, jitter and throughput.

Answer

Latency: time for a packet to travel from source to destination. **Jitter**: variation in that delay between packets of a flow. **Throughput**: amount of data actually delivered per second.

PRACTICE Q4 MEDIUM **[6 marks]**

Use Dijkstra's algorithm to find the shortest paths from S. Links: S–P 4, S–Q 1, Q–P 2, P–R 1, Q–R 5, R–T 3, Q–T 8. Show the table and S's forwarding table.

Answer

Working

Step	N'	P	Q	R	T
0	S	4, S	**1, S**	∞	∞
1	S Q	**3, Q**		6, Q	9, Q
2	S Q P			**4, P**	9, Q
3	S Q P R				**7, R**
4	S Q P R T				

$D(P) = \min(4,\ 1+2) = 3$ ← *via Q beats the direct link*

$D(R) = \min(6,\ 3+1) = 4$; $D(T) = \min(9,\ 4+3) = 7$

Forwarding table of S: P via Q (3), Q via Q (1), R via Q (4, path S–Q–P–R), T via Q (7, path S–Q–P–R–T).

PRACTICE Q5 MEDIUM [6 marks]

Router X has neighbours Y (link cost 1) and Z (link cost 4). Y advertises {X 1, Y 0, Z 2, W 6} and Z advertises {X 4, Y 2, Z 0, W 1}. Compute X's new distance vector and next hops.

Answer

Working

$$D_X(Y) = \min(1+0,\ 4+2) = 1 \text{ via Y}$$
$$D_X(Z) = \min(1+2,\ 4+0) = 3 \text{ via Y} \quad \leftarrow \textit{direct link 4 is worse}$$
$$D_X(W) = \min(1+6,\ 4+1) = 5 \text{ via Z}$$

X's vector: {X 0, Y 1, Z 3, W 5}; next hops Y, Y, Z.

PRACTICE Q6 MEDIUM [5 marks]

A host is shaped by a token bucket of capacity 400 KB filled at 4 MB/s. Its line can send at 20 MB/s. How long can it send at full speed when the bucket starts full, and how much data does that burst carry?

Answer

Working

$$S = \frac{b}{M-r} = \frac{400\ \text{KB}}{(20-4)\ \text{MB/s}}$$
$$= \frac{0.4}{16} = 0.025\ \text{s} = 25\ \text{ms}$$
$$\text{burst} = M \times S = 20 \times 0.025 = 0.5\ \text{MB} \quad \leftarrow \textit{400 KB saved + 100 KB new}$$

After the burst the host sends at the token rate, 4 MB/s.

PRACTICE Q7 MEDIUM [6 marks]

Explain how OSPF builds its routing table and why it uses areas.

Answer

(1) Routers send Hellos to form adjacencies (DR/BDR on LANs). (2) Each router floods LSAs describing its links and their cost. (3) All routers in an area build an identical link-state database.

(4) Each runs Dijkstra SPF with itself as root to get a shortest-path tree, and installs the best routes (cost = $10^8/\text{BW}$). (5) Changes trigger new LSAs only, so convergence is fast. **Areas**: LSAs stay inside an area, so LSDBs are smaller, SPF runs less often and faults are contained; all areas connect to backbone area 0.

PRACTICE Q8 HARD **[10 marks]**

With the aid of a diagram, explain the count-to-infinity problem in distance-vector routing and describe three techniques used to solve it.

Answer

Diagram: R1–R2–R3–Net N, cost 1 per hop, as in Section 4. Before the failure R2 reaches N in 2 hops and R1 in 3. **Problem**: the R2–R3 link fails, so R2 loses its route to N. Before R2 can tell R1, R1 advertises "N in 3". R2 does not know that R1's route goes through R2 itself, so it installs N = 4 via R1. R1 then hears "4" from R2 and raises its cost to 5, R2 goes to 6, and so on. A **routing loop** forms: packets for N bounce between R1 and R2 while the count creeps up to infinity (16 in RIP). Bad news travels slowly because DV routers only know what neighbours tell them. **Solutions** (2 marks each): (1) **Split horizon**: R1 learned N from R2, so it never advertises N back to R2. (2) **Route poisoning + poison reverse**: R3 advertises N with metric 16 at once; neighbours echo it back with 16. (3) **Hold-down timers**: ignore worse news about a failed route for 180 s. Also: **triggered updates** and a **maximum hop count** of 15.

PRACTICE Q9 HARD **[12 marks]**

A company must connect its head office in Harare to branches in Bulawayo and Mutare. Compare a leased line, Frame Relay, MPLS and an Internet VPN for this, and recommend one with reasons. Include a diagram.

Answer

Diagram: like the WAN figure in Section 10, but with three sites, each with its own LAN, router and access link into one provider cloud.

	Cost	Performance / QoS	Comment
Leased line	Highest; one line per pair of sites (3 lines for full mesh)	Guaranteed, private	Does not scale
Frame Relay	Medium; one access line with PVCs	CIR guaranteed; no error correction	Legacy, being withdrawn
MPLS	Medium; any-to-any through one access link	Label switching, QoS classes, SLA	Carrier standard
Internet VPN	Lowest	Best effort; IPsec gives privacy	Good backup

Recommendation: **MPLS** as the main WAN: any-to-any connectivity, QoS for voice and ERP, an SLA from the provider, and easy to add sites; with an **IPsec VPN over broadband** as a cheap backup link. Run PPP or Ethernet on the access links and OSPF/EIGRP between sites.

PRACTICE Q10 HARD **[10 marks]**

(a) Calculate the EIGRP metric of a path whose slowest link is a T1 (1544 kbps) and whose total delay is 20 100 μs. (b) Router R has three routes to 10.5.0.0/16: via B (link 10, RD 20), via C (link 10, RD 25), via D (link 5, RD 35). Identify the successor and any feasible successor.

Answer

Working

$$\text{(a)}\ \frac{10^7}{1544} = 6476 \qquad \leftarrow \textit{truncate}$$

$$\frac{20\,100\ \mu s}{10} = 2010 \qquad \leftarrow \textit{tens of } \mu s$$

$$\text{metric} = 256 \times (6476 + 2010)$$

$$= 256 \times 8486 = 2\,172\,416$$

$$\text{(b)}\ FD_B = 10 + 20 = 30 \qquad \leftarrow \textit{FD = link + RD}$$

$$FD_C = 10 + 25 = 35, \quad FD_D = 5 + 35 = 40$$

$$\text{successor} = \textbf{B}\ (\text{lowest FD} = 30)$$

$$RD_C = 25 < 30 \Rightarrow \textbf{C is feasible successor}$$

$$RD_D = 35 \not< 30 \Rightarrow \text{D is not} \qquad \leftarrow \textit{might loop via R}$$

If B fails, R switches to C immediately without recomputing; D would only be used after DUAL queries.

PRACTICE Q11 HARD **[12 marks]**

A hospital network carries VoIP, video consultations, patient-record database traffic and staff web browsing over one congested WAN link. Design a QoS policy for it.

Answer

1. **Classify and mark** at the edge with DiffServ: VoIP = EF (46), video = AF41, database = AF31, web = best effort (0).
2. **Queuing**: low-latency (strict priority) queue for voice, capped at about 30% of the link so it cannot starve others; class-based weighted fair queuing for video and database; web gets what is left.
3. **Shaping** / **policing**: token bucket on video and web so bursts do not fill queues; police web downloads.
4. **Congestion avoidance**: WRED on the data queues so TCP flows slow down early instead of tail drop.
5. **Admission control**: refuse a new video call if the video class is full rather than degrade every call.
6. **Targets**: voice one-way delay < 150 ms, jitter < 30 ms, loss $< 1\%$; monitor with SNMP and get them written into the carrier's SLA. Long-term: add bandwidth or a second link.

ONE-PAGE CHEAT SHEET

Routing table sources: Connected (C, AD 0), Static (S, 1), Dynamic: EIGRP 90, OSPF 110, RIP 120. Lowest AD wins.
Classes: flooding, static (non-adaptive), dynamic (adaptive). **Goals**: optimality, simplicity, robustness, rapid convergence, flexibility. **Metrics**: hop count, reliability, delay, bandwidth, load, cost.
DV vs LS

- DV: whole table, neighbours, periodic, Bellman-Ford, slow, loops (RIP)
- LS: LSAs, flood to all, on change, Dijkstra, fast, more CPU (OSPF)

Table 1: RIPv1 DV hop count (max 15); EIGRP adv. DV bandwidth + delay; OSPF LS cost = $10^8/\text{BW}$.
Dijkstra: add the cheapest node, update $D(v) = \min(D(v), D(w) + c(w, v))$, keep predecessor.
Bellman-Ford: $D_x(y) = \min_v\{c(x, v) + D_v(y)\}$.
Count-to-infinity fixes: max 15 hops, split horizon, route poisoning, poison reverse, hold-down, triggered updates.
RIPv2: 30 s full table, 224.0.0.9, UDP 520, classless, MD5; timers 30/180/180/240.
EIGRP: hellos 224.0.0.10, 3 tables, DUAL; successor = lowest FD; feasible successor if RD $<$ FD.
Congestion: load $>$ capacity $\Rightarrow$ delay, drops. Open loop (prevent): retransmission, window, ACK, discard policies, admission control, shaping. Closed loop (remove): backpressure, choke packet, implicit/explicit (ECN) signalling, RED.
Leaky bucket: constant output r, drops packets. **Token bucket**: bursts up to b, drops tokens, $S = b/(M - r)$.
QoS parameters: bandwidth/throughput, latency, jitter, packet loss (+ reliability). **Best effort**: no guarantees, all equal, drop on congestion. **VoIP**: low BW, delay $<$ 150 ms, jitter $<$ 30 ms, loss $<$ 1%, EF. **Video**: high BW, bursty, streaming buffers. IntServ (RSVP, per flow) vs DiffServ (DSCP classes).
SDN: control plane separated and centralised in a controller; northbound API (apps), southbound API (OpenFlow) to switches.
ISP tiers: T1 backbone, peer free; T2 regional, buy transit, IXPs; T3 local access. BGP between ASes.
SLA: contract; uptime, bandwidth, latency/jitter/loss, repair times, penalties.
WAN: leased line, PPP (LCP/NCP, PAP/CHAP), HDLC (flag 01111110), X.25, Frame Relay (DLCI, PVC), ATM (53-B cells), MPLS (labels, QoS, VPN).